

5. Test Environment Setup

Overview

Before executing API test cases, it is essential to set up the testing environment properly. A well-configured test environment ensures smooth execution of API requests, proper management of test data, successful automation, and accurate report generation.

In this project, the test environment consists of Postman, Fake Store API, Environment Variables, Newman CLI, and Node.js. These tools work together to create, execute, validate, and automate API tests.

The environment setup process includes:

- Installing Postman
- Creating a Postman Collection
- Configuring Environment Variables
- Installing Node.js
- Installing Newman CLI
- Configuring Newman Report Generation

The following sections explain each setup activity in detail.

5.1 Postman Installation

Purpose

Postman is the primary tool used in this project for API testing. It allows testers to create requests, send HTTP requests, validate responses, write test scripts, and organize APIs into collections.

Installation Steps

Step 1: Download Postman

Visit the official Postman website:

<https://www.postman.com/downloads/>

Step 2: Install Postman

- Download the installer based on your operating system.
 - Run the installation file.
 - Follow the installation wizard.
 - Complete the installation process.
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Step 3: Launch Postman

After installation:

- Open Postman.
- Sign in using your account (optional).
- Create a workspace for API testing.

Example Workspace:

`FakeStore_API_Testing`

Verification

To verify installation:

- Open Postman successfully.
 - Create a new request.
 - Send a sample GET request.
 - Verify that a response is received.
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5.2 Environment Configuration

Purpose

Environment configuration helps manage reusable values across multiple API requests.

Instead of hardcoding URLs and IDs in every request, environment variables are used.

Benefits:

- Easy maintenance
 - Reduced duplication
 - Faster execution
 - Better project organization
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Environment Creation Steps

Step 1

Open Postman.

Step 2

Click:

Environments

Step 3

Click:

Create Environment

Step 4

Enter Environment Name:

FakeStore_Environment

Step 5

Save the Environment.

Environment Usage

The environment is selected before executing API requests.

Example:

```
Environment:  
FakeStore_Environment
```

This ensures all requests use the correct variables during execution.

5.3 Collection Creation

Purpose

A Collection is used to organize all API requests in a structured manner.

Collections improve maintainability and allow multiple APIs to be executed together using Collection Runner and Newman.

Collection Creation Steps

Step 1

Open Postman.

Step 2

Click:

New Collection

Step 3

Provide Collection Name:

FakeStore_API_Collection

Step 4

Create folders inside the collection.

Collection Structure

FakeStore_API_Collection

```
|— Authentication
|   |— Login User
```

```
|— Products
|   |— Get All Products
|   |— Get Single Product
```

	└─ Add New Product
	└─ Update Product (PUT)
	└─ Update Product (PATCH)
	└─ Delete Product
└─ Categories	
	└─ Get All Categories
	└─ Get Products in Category
└─ Cart	
	└─ Get All Carts
	└─ Get Single Cart
	└─ Add to Cart
	└─ Update Cart (PUT)
	└─ Update Cart (PATCH)
	└─ Delete Cart
└─ Users	
	└─ Get All Users
	└─ Get Single User

Benefits

- Better organization
 - Easy navigation
 - Faster execution
 - Supports Collection Runner
 - Supports Newman automation
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5.4 Environment Variables Setup

Purpose

Environment variables are used to store reusable values that can be accessed across multiple requests.

This reduces hardcoding and makes the project easier to maintain.

Variables Created

Base URL

Variable Name:
baseUrl

Value:
https://fakestoreapi.com

Product ID

Variable Name:
productId

Value:
1

User ID

Variable Name:
userId

Value:
1

Cart ID

Variable Name:
cartId

Value:
1

Authentication Token

Variable Name:
token

Value:
Generated Dynamically

Example Usage

Instead of writing:

`https://fakestoreapi.com/products`

Use:

`{{baseUrl}}/products`

Benefits:

- Easy maintenance
- Reusable configuration
- Supports multiple environments

5.5 Newman Setup

Purpose

Newman is Postman's command-line collection runner used to automate API execution and generate reports.

Newman enables execution of collections without opening Postman.

Prerequisites

Before installing Newman:

- Node.js must be installed.
 - NPM must be available.
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Verify Node.js Installation

Open Command Prompt:

```
node -v
```

Expected Output:

```
vXX.X.X
```

Verify NPM Installation

```
npm -v
```

Expected Output:

```
XX.X.X
```

Install Newman

Run:

```
npm install -g newman
```

Verify Newman Installation

```
newman -v
```

Expected Output:

```
Newman Version Number
```

Install Newman HTML Reporter

```
npm install -g newman-reporter-htmlextra
```

This package is used to generate detailed HTML reports after collection execution.

Execute Collection Using Newman

```
newman run FakeStore_API_Collection.json
```

Execute Collection with Environment

```
newman run FakeStore_API_Collection.json -e FakeStore_Environment.json
```

Generate HTML Report

```
newman run FakeStore_API_Collection.json -e FakeStore_Environment.json -r cli,htmlextra
```

Expected Output

After successful execution:

- Total Requests Executed
 - Total Assertions Executed
 - Passed Tests
 - Failed Tests
 - Response Time Statistics
 - HTML Report Generated
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Test Environment Summary

The test environment for this project was configured using Postman, Fake Store API, Environment Variables, Node.js, and Newman CLI. Postman was used for API creation and validation, environment variables were configured for reusable test data, collections were organized for efficient execution, and Newman was integrated to automate execution and generate detailed HTML reports. This setup provides a complete and industry-standard environment for API testing and automation.